

Problem Statement – Part 1

<https://github.com/vishalanand49/AI-ML-Project>

Assignment-based Subjective Questions

1. From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable?

Analysis of the categorical variables from the dataset is as below:

- a) Seasonality: Bike rental demand is highest during the summer and fall seasons and tends to drop during winter.
- b) Weather Impact: People are more likely to rent bikes when the weather is clear and pleasant. Demand decreases during rainy or snowy conditions.
- c) Yearly Trend: Bike rentals were higher in 2019 compared to 2018, indicating that bike-sharing services became more popular over time.
- d) Working Days: Usage is generally higher on working days, suggesting that many riders use bikes for their daily commute.

2. Why is it important to use `drop_first=True` during dummy variable creation?

Using `drop_first=True` removes one category from each variable.

If you don't do this, the variables become too similar (highly correlated), which can confuse the model and make results unreliable.

3. Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable?

Temperature shows strong positive linear relationship with `cnt`

4. How did you validate the assumptions of Linear Regression after building the model on the training set?

After building the model, we did some basic checks:

- a) Linearity: Compared predicted vs actual values using a scatter plot
- b) Normality: Looked at residuals using a histogram → they should follow a normal pattern

5. Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes?

Top 3 features contributing significantly towards explaining the demand of the shared bike are:

- a) Temperature
- b) Year
- c) Weather

General Subjective Questions

1. Explain the linear regression algorithm in detail

Linear Regression is used to predict a number (like price, sales, etc.) based on other inputs.

- a) It finds a best-fit line between input (X) and output (Y)
- b) The goal is to make predictions as close as possible to actual values
- c) It does this by reducing the error (difference between actual and predicted)
- d) It also tells how much each input affects the output

2. Explain the Anscombe's quartet in detail.

All four datasets may have the same mean, variance, and regression line, which means statistically they look identical.

However, when you visualize the data (using graphs), you'll notice that each dataset has a completely different pattern.

3. What is Pearson's R?

Pearson's R - Measures linear relationship between two variables.

4. What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling?

Scaling is adjusting data to a similar range which prevents data with higher values dominating the model

Normalized scaling: Fixed range 0 and 1

Standardized scaling: Centered around the average

5. You might have observed that sometimes the value of VIF is infinite. Why does this happen?

The value of VIF (Variance Inflation Factor) becomes infinite when there is perfect multicollinearity among independent variables.

It happens when one variable is an exact linear combination of others

6. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression.

A Q-Q plot is an essential diagnostic tool used to verify whether residuals follow a normal distribution, which is critical for the accuracy and reliability of a linear regression model.